

High-Growth Companies

Valuing high-growth companies is a challenge; some practitioners even describe it as hopeless. Yet we've found that the valuation principles in this book work well for coping with the great uncertainty that accompanies these rapid growers.¹ The best way to value such companies is to start with a discounted-cash-flow (DCF) valuation and buttress it with economic fundamentals and probability-weighted scenarios.

Although DCF may sound suspiciously retro, it works where other methods fail, since the core principles of economics and finance apply even in uncharted territory. Alternatives, such as enterprise value multiples, generate imprecise results when earnings are highly volatile, cannot be used when earnings are negative, and provide little insight into what drives the company's valuation. More important, shorthand methods cannot account for the unique characteristics of each company in a fast-changing environment. Another alternative, real options, requires estimates of the long-term revenue growth rate, long-term volatility of revenue growth, and profit margins—the same requirements as for discounted cash flow.²

This chapter details the differences in the order and emphasis of DCF valuation in the case of high-growth, rather than established, companies. Instead of starting with an analysis of the company's and its industry's past performance, the valuation process begins with an estimation of what the future economics of the company and industry might become. Since these long-term projections are highly uncertain, create multiple scenarios, each with its own value. If you need a single-point estimate, weight the scenario values by their probability of occurrence. In our practice, we avoid using single-point

¹ We define high-growth companies as those whose organic revenue growth exceeds 15 percent annually.

² In Chapter 39, we demonstrate how real options can lead to a more theoretically robust valuation than scenario analysis. But unlike scenario analysis, real-options models are complex and obscure the competitive dynamics driving a company's value.

estimates, because this approach implies precision that doesn't exist and obscures the key uncertainties that could improve decision making.

Keep in mind that while scenario-based DCF techniques can help bound and quantify uncertainty, they will not make it disappear. High-growth companies have volatile stock prices for sound reasons.

A VALUATION PROCESS FOR HIGH-GROWTH COMPANIES

When valuing an established company, the first step is to analyze historical performance. But in the case of a high-growth company, historical financial results provide limited clues about future prospects. Therefore, begin with the future, not with the past. Focus on sizing the potential market, estimating the share of the market the company will capture, predicting the level of sustainable operating margin, and approximating the investments necessary to achieve scale. To make these estimates, choose a point well into the future at a time when the company's financial performance is likely to stabilize, and begin forecasting.

Once you have developed a long-term future view, work backward to link the future to current performance. Current performance measured using accounting statements will mix together investments and expenses. When possible, capitalize hidden investments, even those expensed under traditional accounting rules.³ This is challenging, as the distinction between investment and expense is often unobservable and subjective.

Given the uncertainty associated with high-growth companies, do not rely on a single long-term forecast. Describe the market's development in terms of multiple scenarios, including total size, likely competitive structure, and so on. When you build a comprehensive scenario, be sure all forecasts, including revenue growth, profitability margins, and required investment, are consistent with the underlying assumptions of the particular scenario. Apply probabilistic weights to each scenario, using weights that are consistent with long-term historical evidence on corporate growth. As we saw during the dot-com bubble of the late 1990s, valuations that rely on unrealistic assessments can lead to overestimates of value, poor investment returns, and strategic errors.

Start from the Future

Begin by thinking about what the industry and company might look like as the company evolves from its current high-growth, uncertain condition to a

³ Chapter 24 presents a methodology for capitalizing intangible expenses, such as research and development.

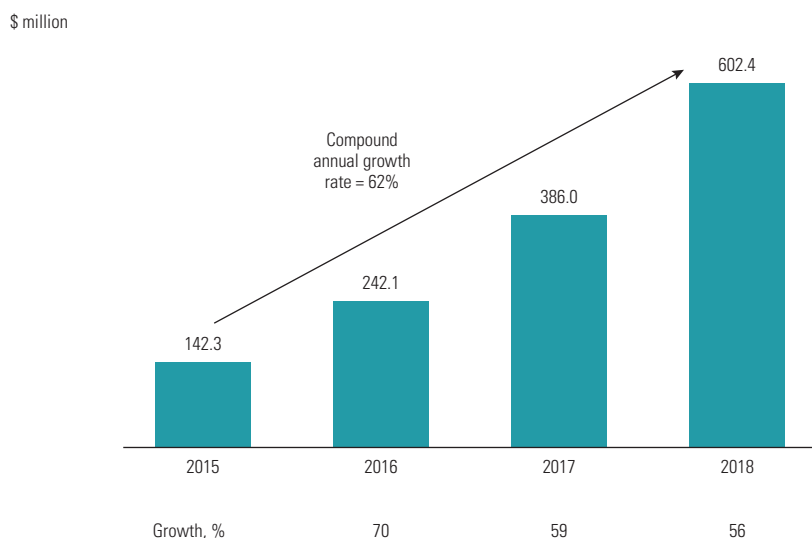
state of sustainable, moderate growth in the future. Then interpolate back to current performance. The future state should be defined and bounded by measures of operating performance, such as customer penetration rates, average revenue per customer, and sustainable margins. Next, determine how long growth will continue at an elevated rate before it stabilizes to normal levels. Since most high-growth companies are start-ups, stable economics probably lie at least 10 to 15 years in the future.

To demonstrate the valuation process for high-growth companies, we examine Farfetch, a popular online marketplace for luxury goods. Founded in 2007 by José Neves, Farfetch was conceived in response to the founder's own struggles to transition from an in-store boutique to a web-based e-tailer. Neves believed that local boutiques lacked the skills and scale to successfully migrate the transition to digital on their own. Farfetch would fill this gap.

The company launched its website in 2008, selling luxury products from 25 boutiques in five countries. Over the next decade, the company raised nearly \$700 million in private capital, growing revenue to more than \$600 million by 2018. As Exhibit 36.1 demonstrates, the company has grown at more than 50 percent in each of the last three years. This level of growth significantly outpaces the growth at more established technology-enabled marketplaces.

To estimate the size of a potential market, start by assessing how the company fulfills a customer need. Then determine how the company generates

EXHIBIT 36.1 **Farfetch: Revenues, 2015–2018**

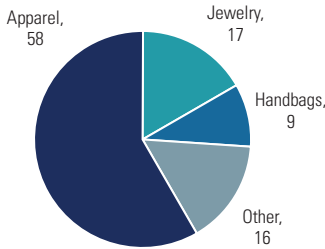


Source: Farfetch F-1 filing and 2018 20-F filing.

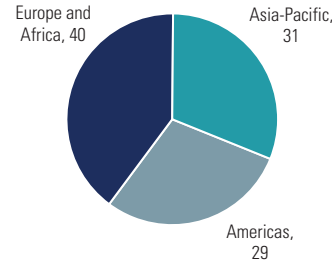
EXHIBIT 36.2 **Farfetch: Revenue by Product Type and Geography**

%

Revenue by type, Q2 2018



Revenue by geography, FY 2018



Source: Farfetch F-1 filing and 2018 20-F filing; Deutsche Bank estimates.

(or plans to generate) revenues. Understanding how a start-up makes money is critical. Technology start-ups rely on many revenue streams, including advertising, product sales, subscriptions, and commissions, among others. Many young companies build a product or service that meets the customer's need, but too many can't identify how to monetize the value they provide.

Understanding a company's growth potential requires identifying which product categories are part of its current and future portfolio. To this end, the left side of Exhibit 36.2 presents Farfetch revenue by product type. While high-end fashion apparel accounts for the majority of its sales, the company also sells high-end jewelry, handbags, and shoes.

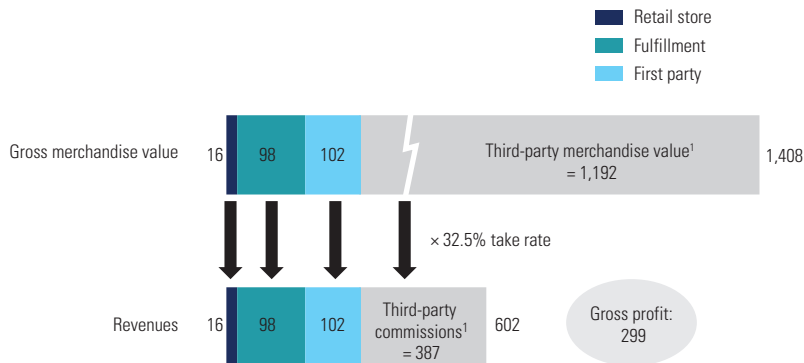
In the case of luxury goods, it is important to assess where the company sells its products, since the luxury-goods market varies dramatically across regions. Understanding the geographic presence will help with sizing future markets and assessing the impact of potential competition. The right side of Exhibit 36.2 presents Farfetch's revenue by geography. Although Farfetch launched in Europe, it now has a significant presence in the Americas and Asia.

Across these product lines and regions, Farfetch generates revenue from multiple activities. Whenever possible, try to separate sources of revenue, as each will have its own dynamics concerning growth, profitability, and required investment. Farfetch's primary source of revenue is from its third-party (3P) marketplace. As in other popular marketplaces, a consumer purchases a product from a company other than Farfetch, and Farfetch facilitates the transaction, taking a portion of the revenue. In a technology-enabled marketplace, the level of the commission is known as the "take rate," and it varies substantially across product categories. For Farfetch, the take rate hovers around 30 percent, higher than most technology marketplaces. As part of the transaction,

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EXHIBIT 36.3 Farfetch: Revenue Model, 2018

\$ million



¹ Includes Black & White outsourcing revenue, estimated net of returns.

Source: Farfetch F-1 filing and 2018 20-F filing, Cowen and Company estimates.

Farfetch will also charge for shipping, customs, and taxes. While these fees are bundled together from the customer's perspective, Farfetch separates fulfillment-related charges from other revenues.

In addition to the marketplace, Farfetch generates revenue from three other sources. The company sells luxury goods direct to consumers through its platform (first-party sales, or 1P) and through two London-based retail stores, known as Browns. Through a business unit it calls Black & White (now known as Farfetch Platform Solutions), Farfetch also works directly with luxury brands to operate their e-commerce sites.

Since a company's take rate varies over time and across businesses, do not start your valuation with company revenue, but rather with gross merchandise value (GMV). Exhibit 36.3 presents Farfetch's GMV and resulting revenue by operating segment. GMV represents the value of goods sold on the platform, net of returns—\$1.4 billion in 2018. Since Farfetch keeps only a portion of the gross merchandise value traded, revenue is limited to the portion retained. This is not unique to Farfetch. Many technology companies report both gross and net sales. For instance, ride-sharing companies report their gross bookings but net out driver payments before reporting revenue. Assess the market power of various stakeholders, like luxury boutiques or global luxury brands, to determine the future direction of take rates.

For in-store sales, first-party sales, and platform fulfillment, revenue equals the gross merchandise value. Since direct sales and the third-party marketplace have different levels of profitability and capital needs, always analyze them separately. In this chapter we examine only the third-party marketplace in detail, though we estimated the other revenue sources using a similar methodology.

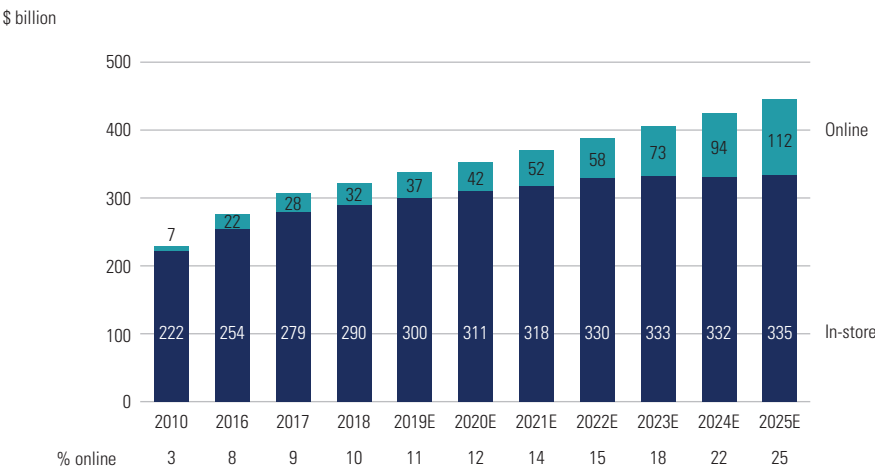
Sizing the Market For many young, high-growth companies, it's challenging to estimate the size of the potential market. You must be creative and clever about what data to collect to bound your forecasts. In the case of the ride-share market, for example, you can start with global taxi revenue. But you must also consider how many new rides will occur because of the ease of access, as well as how many riders are likely to replace their cars altogether.

Compared with other nascent technology companies, Farfetch's product markets are easier to size. Many investment banks and consulting firms study the global luxury market in depth. Information is available on the number of products and the total value of merchandise sold in stores and online.

Exhibit 36.4 presents revenue for the global luxury market between 2010 and 2018, with analyst forecasts extending through 2025. Bolstered by strong growth in Asia, the luxury market is expected to grow between 4 and 5 percent annually, reaching \$450 billion by 2025. Growth looks to be uneven across channels, however. Expectations are that over the next decade, in-store sales of luxury products will remain flat, while online sales will increase from 10 percent to 25 percent of the total market.

The online migration of luxury goods is far from certain, however. No one really knows if these high-end products will migrate as much or as quickly as have media, electronics, and other items. Later in the chapter we develop multiple scenarios, with each scenario based on a different level of online penetration.

EXHIBIT 36.4 **Global Luxury Market: Offline vs. Online Sales, 2010–2025E**



Source: Farfetch Investor Presentation, December 4, 2019, Bain & Company.

EXHIBIT 36.5 **Farfetch: Revenue Model for Third-Party Marketplace, 2017–2028E**

\$ million, except where noted

	2017	2018	2019E	2020E	2021E	2022E	...	2028E
Third-party revenue model								
Unique customers, thousands	936	1,353	1,787	2,293	2,897	3,603	...	9,911
× Orders per customer	2.01	2.15	2.31	2.33	2.35	2.38	...	2.52
Number of orders, thousands	1,881	2,913	4,121	5,342	6,817	8,563	...	25,005
× Average order value, \$	620	618	605	637	650	659	...	700
Gross transaction value ¹	1,166	1,801	2,493	3,402	4,428	5,645	...	17,500
Net of returns	(290)	(439)	(577)	(782)	(1,018)	(1,411)	...	(4,375)
Net of fulfillment revenue	(74)	(98)	(138)	(197)	(256)	(318)	...	(984)
GMV, net of fulfillment revenue ²	802	1,265	1,777	2,423	3,154	3,917	...	12,140
× Portion generated by third party	95.0%	91.9%	91.5%	92.2%	92.8%	93.3%	...	93.3%
× Take rate	32.8%	32.5%	30.9%	30.9%	30.6%	30.3%	...	29.4%
Third-party revenue	250	377	503	691	896	1,108	...	3,331
Other Revenue								
Platform fulfillment revenue	74	98	138	197	256	318	...	984
First-party revenue	40	102	152	189	228	261	...	815
Black & White segment	6	10	16	23	30	37	...	135
Browns' in-store revenue	15	16	18	20	22	23	...	30
Farfetch revenue	386	602	827	1,119	1,430	1,746	...	5,296

¹ Excluding Black & White's outsourcing revenue and Browns' in-store revenue.² Gross merchandise value, net of fulfillment revenue.

Source: Farfetch F-1 filing and 2018 20-F filing; Cowen and Company estimates.

Modeling Revenues A robust revenue model, especially for nascent companies, will incorporate operational data mirroring the company's economics. In the case of consumer companies, these include the number of customers, transactions per customer, average transaction size, and other items. Exhibit 36.5 presents a revenue model for Farfetch's third-party marketplace, based on forecasts by the research team at Cowen and Company.⁴ These particular estimates underpin one of our four scenarios presented later in the chapter.

The revenue model in Exhibit 36.5 starts with the number of customers, which in 2018 was approximately 1.4 million. Since the typical customer makes a purchase just over twice a year, the company generated just under 3 million transactions that year. The size of the average order was about \$600, leading to an estimated \$1.8 billion in gross transaction value. Since some transactions will

⁴ To create one of our scenarios, we used data from John Blackledge et al., "FTCH 2Q Preview: Order Growth Momentum in Focus when FTCH Reports 8/8" (unpublished manuscript, Cowen and Company, August 5, 2019), available from Bloomberg. We thank the team at Cowen and Company for providing data used throughout the chapter.

be returned, they are netted from the original order count. We also net out process fulfillment costs, leading to GMV, net of fulfillment revenue of \$1.3 billion—just under 4 percent of the online luxury market, estimated at \$32 billion.

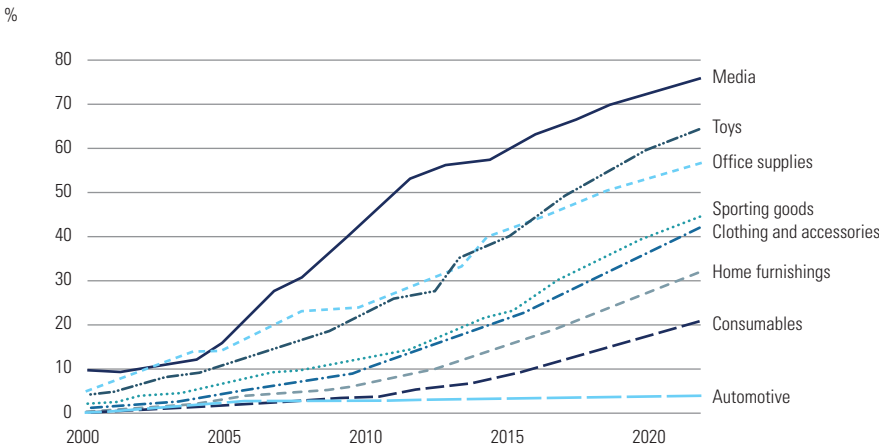
We assume that 2028 is the year that Farfetch matures into a profitable, stable company, so we create our projections for this scenario around estimates for that year’s results. We project 9.9 million unique customers and 2.5 transactions per customer to arrive at 25 million transactions at an average of \$700 per transaction. That leads to gross transaction value near \$17.5 billion. After returns and adjustment for fulfillment costs, revenues from Farfetch’s digital platform would be \$12.1 billion.

With 93.3 percent of revenue generated from third-party sales and a 29.4 percent take rate, third-party revenues equal \$3.3 billion. First-party revenues equal \$815 million (that is, 6.7 percent of \$12.1 billion). Other revenues, primarily from process fulfillment, outsourcing, and in-store sales, are \$1.1 billion, for total revenues of \$5.3 billion.

We also looked at this revenue estimate from the perspective of the total market and Farfetch’s share. The total luxury-goods market, growing modestly, reaches about \$500 billion in 2028. Assuming 30 percent online penetration and Farfetch achieving an 8 percent market share (with adjustments for first-party versus third-party sales and in-store sales) brings us to a scenario in which Farfetch achieves 2028 revenue of \$5.3 billion.

One of the key uncertainties for estimating Farfetch’s revenues is online penetration of the luxury-goods market. For this scenario, our projected 30 percent penetration is based on a comparison with other product categories. Exhibit 36.6

EXHIBIT 36.6 U.S. E-Commerce Penetration by Vertical, 2000–2023E



Source: U.S. Census Bureau; Cowen and Company estimates.

presents e-commerce penetration in the United States across various product categories. Most sales of media and toy products now take place online. In contrast, luxury goods lag most product categories. One product category that analysts point to as a useful comparable is clothing and accessories, projected to reach 42 percent by 2023. Luxury-goods customers like to feel and touch their purchases, so for this scenario, we've kept penetration at 30 percent. In the most optimistic scenario we'll use for our valuation, we'll bring penetration to 40 percent.

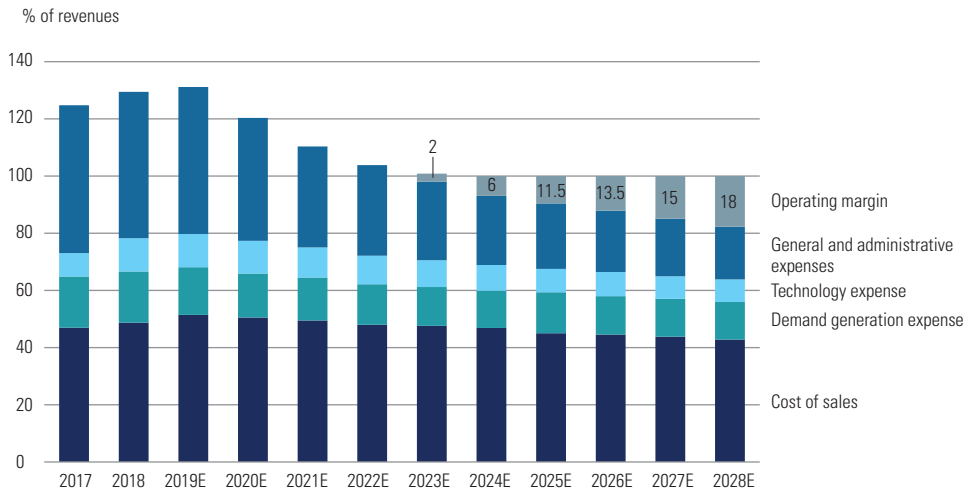
Estimating Operating Margin, Capital Intensity, and ROIC With a revenue forecast in hand, next forecast long-term operating margins, required capital investments, and ROIC. To estimate operating margin, triangulate between expected price versus cost to serve and operating margins for established players. Refer to Chapter 8 for the range of ROIC for different industries and a discussion of the drivers of ROIC. Chapter 6 discusses the types of business conditions that lead to high levels of ROIC from network effects.

Because the underlying economics of Farfetch's business segments differ so much, it is important to evaluate each segment separately. Regarding the company's third-party marketplace, Farfetch neither manufactures nor holds inventory, but only facilitates transactions between other parties. Therefore, the segment has a low cost of sales, requires little capital, and is highly scalable. To project steady-state margins and capital intensity, we look to established players that operate technology marketplaces that earned margins exceeding 20 percent. While these competitors can provide some insight on the evolution of margins and capital intensity, differences in each competitor's business mix may hamper direct comparison. Therefore, substantial judgment is required.

In the case of first-party sales directly to the consumer, margins will more closely resemble those of an online e-tailer (for sales conducted online) or a luxury-goods retailer (for sales conducted in their stores). In contrast to the third-party marketplace, growth will require purchasing more goods, holding more inventory, and perhaps someday capital investment in warehouses and stores. These margins are much lower than for technology marketplaces, often far less than 10 percent.

Exhibit 36.7 presents one research analyst's forecast of how Farfetch's margins might evolve over time. Although gross profits were positive in 2018, the company was losing money, as marketing and other support expenses exceeded the gross profits from product sales. Technology costs and general expenses are expected to rise as the company grows, though at a slower rate than revenue. As a consequence, these expenses will drop as a percentage of sales, and operating margins will increase. This is not the case for every expense. Some accounts, such as cost of sales, will remain a relatively constant

EXHIBIT 36.7 Farfetch: Current and Forecast Margins, 2017–2028E



Source: Farfetch F-1 filing and 2018 20-F filing; Cowen and Company estimates.

proportion of sales. This is because the company will need to purchase additional products to support higher sales.

For 2028, the exhibit shows a forecast operating profit margin of 18 percent, which we'll use in our scenario B. Later, we'll show a range of margin forecasts. We've also assumed that Farfetch's capital productivity is a hybrid of a marketplace and e-tailer in proportion to Farfetch's relative third-party versus first-party sales.

Work Backward to Current Performance

After completing a forecast for total market size, market share, operating margin, and capital intensity, reconnect the long-term forecast to current performance. To do this, you must assess the speed of transition from current performance to future long-term performance. Estimates must be consistent with economic principles and industry characteristics. For instance, from the perspective of operating margin, how long will fixed costs dominate variable costs, resulting in low margins? Concerning capital turnover, what scale is required before revenues rise faster than capital? As scale is reached, will competition drive down prices? Often the questions outnumber the answers.

To determine the speed of transition from current performance to target performance, examine the historical progression for similar companies. Unfortunately, analyzing historical financial performance for high-growth companies is often misleading, because long-term investments for high-growth companies tend to be intangible. Under current accounting rules, these

investments must be expensed. Therefore, both early accounting profits and invested capital will be understated. With so little formal capital, many companies have unreasonably high ROICs as soon as they become profitable.

Develop Scenarios

A simple and straightforward way to deal with uncertainty associated with high-growth companies is to use probability-weighted scenarios. Developing even a few scenarios makes the critical assumptions and interactions more transparent than you will achieve with other modeling approaches, such as real options and Monte Carlo simulation.

To develop probability-weighted scenarios, estimate financial performance for a full range of outcomes, some optimistic and some pessimistic. For Farfetch, we have developed four future scenarios for 2028, summarized in Exhibit 36.8.

In scenario A, we forecast that Farfetch benefits from favorable market conditions and delayed competitive entry. While the aggregate luxury-goods market continues to grow at a steady pace, online penetration in the space accelerates beyond analyst expectations as brands and consumers quickly embrace electronic channels. Online adoption in the luxury-goods market mimics the pattern of the clothing market, peaking at 40 percent penetration. Farfetch achieves 12 percent market share, leading to \$11.4 billion in revenue in 2028. Competitive entry is forestalled, and operating margins approach those of best-in-class technology marketplaces at 22 percent. Scenario A represents an

EXHIBIT 36.8 **Farfetch: Key Drivers by Scenario, 2028 Forecast**

	Online penetration of luxury, %	Farfetch market share, ¹ %	Total revenues, \$ million	Operating margin, %	Description
Scenario A	40	12	11.4	22	Online penetration of the luxury market accelerates, following the path of clothing and accessories; margins are strong as the company leads the category.
Scenario B	30	8	5.3	18	Online penetration follows historical progression; margins match analyst expectations.
Scenario C	30	5	3.3	14	Online penetration follows historical progression; margins fail to materialize as larger competitors enter the category.
Scenario D	15	4	1.2	6	Online penetration fails to meet expectations; margin pressure intensifies due to large-scale entrants and increased omnichannel presence from traditional retailers.

¹ Measured as gross merchandise value to online purchases.

optimistic forecast, but structural similarities between the online luxury and clothing markets make this scenario entirely plausible.

Scenarios B through D follow a similar construct but vary key assumptions. In scenarios B and C, penetration rates reach only 30 percent, reflecting the desire among luxury goods buyers for a greater physical shopping experience than with other categories. Farfetch's market share reaches a healthy, but not overly aggressive, 8 percent and 5 percent, respectively. Margins are somewhat lower than the best e-tailers because of stronger supplier market power, in the range of 14 to 18 percent. Scenario D is characterized by sluggish growth in online penetration that more closely mirrors the home furnishings industry, a segment that is less conducive to electronic retail. In scenario D, by 2028, online penetration is 15 percent, and Farfetch achieves revenues of just \$1.2 billion. Increased pressure from new entrants and more widespread omnichannel adoption by individual brands result in more moderate margin expansion, reaching only 6 percent, comparable to discount retailers.

Weight Scenarios

To derive current equity value for Farfetch, weight the intrinsic equity valuation from each scenario by its estimated likelihood of occurrence, and sum across the weighted scenarios. Exhibit 36.9 lists the intrinsic equity valuations and the probability of occurrence for each scenario. At a 10 percent probability for scenario A, 30 percent for scenario B, 35 percent for scenario C, and 25 percent for scenario D, Farfetch's equity value equals \$6.1 billion and value per share at \$20, matching its 2018 IPO price. Whether this price is appropriate depends on your belief in the forecasts and their respective probabilities. Were they too optimistic, too pessimistic, or just right?

EXHIBIT 36.9 **Farfetch: Probability-Weighted Expected Value**

Scenario	Intrinsic equity valuation, \$ billion	×	Probability, %	=	Contribution to equity valuation, \$ billion
Scenario A	19.6		10		2.0
Scenario B	8.2		30		2.5
Scenario C	4.1		35		1.4
Scenario D	1.0		25		0.2
			100		6.1
Shares outstanding, millions					300.0
Value per share, \$					20

UNCERTAINTY IS HERE TO STAY

By adapting the DCF approach, it is possible to generate reasonable valuations for dramatically changing businesses. But investors and companies entering fast-growth markets like those related to new technologies and complex business ecosystems still face huge uncertainties. To see why, look at what could happen under our four scenarios to an investor who holds a share of Farfetch stock for five years after buying it in 2018 for \$20. To facilitate the calculation, we assume the investor gradually learns about the most likely scenario.

If scenario A plays out, the investor will earn a 39 percent annual return, and as of 2018, the market will seem to have drastically undervalued Farfetch. If scenario C plays out, the investment will make just 2 percent annually, failing to earn its cost of capital. If scenario D plays out, the investment will lose 23 percent a year, and it will appear that the company was significantly overvalued in 2018. Going forward, these high or low potential returns should not be interpreted as implying that the current share price was irrational; they merely reflect uncertainty about the future.

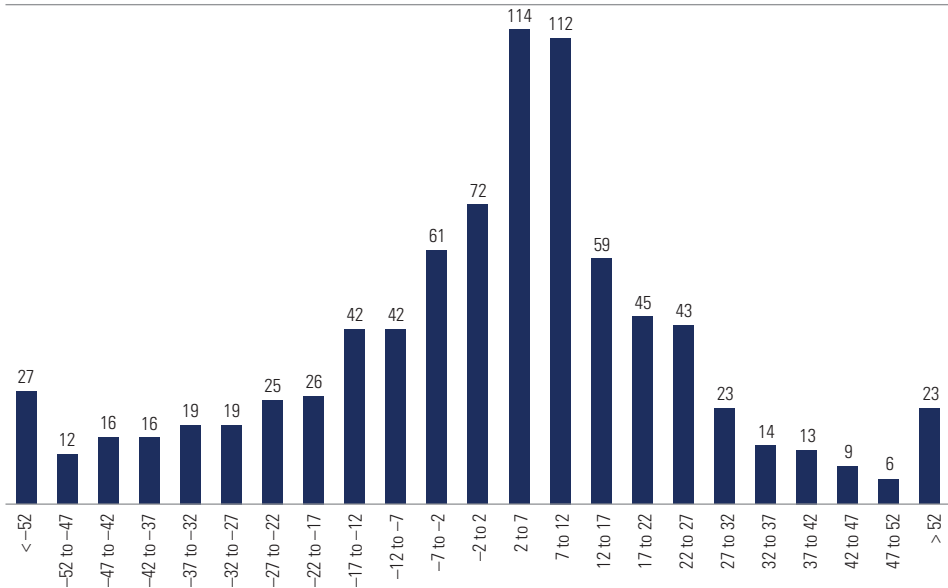
Accurately predicting which scenario will occur is a laudable goal, but unlikely to happen. Investors struggle to incorporate new information every day, and this leads to high volatility in the share prices of young companies. Farfetch, for instance, saw its price rise 50 percent on the day of its initial public offering (IPO). Then its shares dropped 40 percent in August 2019, when

EXHIBIT 36.10 **Farfetch: Share Price, 2018–2019**



EXHIBIT 36.11 **Distribution of Annualized Total Shareholder Returns for U.S. IPOs**

Number of companies



Note: Total shareholder returns for 838 initial public offerings (IPOs) between 2010 and 2017. Returns are measured from the first day of trading through December 31, 2019.

estimates on transaction counts were revised downward (see Exhibit 36.10).⁵ The company had five times the volatility of the S&P 500 during its first two years of trading.

As Farfetch's prospects begin to stabilize, however, it should be possible to tighten the range of potential outcomes. These gains in precision should be reflected in a decrease in the stock's volatility.

The challenge of accurate valuation is not limited to Farfetch. We examined the total shareholder returns for more than 800 initial public offerings since 2010. Only 112 of the 838 IPOs earned between 7 and 12 percent, a range many consider the fair rate of return for investing in equities. Instead, investors either made or lost much more than anticipated. In fact, nearly 10 percent of IPOs either generated or lost 50 percent of their value since going public.⁶

A great deal of uncertainty is associated with the problem of identifying the eventual winner in a competitive field. History shows that a few players

⁵ In August 2019, Farfetch announced the acquisition of New Guards Group, an Italian brand platform that operates a portfolio of luxury fashion labels. The company purchased New Guards to further differentiate its product portfolio and capture a greater share of the online market, but some analysts expressed concern about a potential shift away from the company's asset-light third-party model. At the same time, Farfetch lowered near-term GMV forecasts to reflect a decrease in promotional spending. We believe that our four scenarios, modeled earlier in the year, still ring true, albeit with a greater probability for the less favorable scenarios than when originally created.

⁶ The results come from Corporate Performance Analytics by McKinsey, which relies on financial data provided by Standard & Poor's Compustat and Capital IQ.

will win big, while the vast majority will toil away in obscurity. It is difficult to predict which companies will prosper and which will not. Neither investors nor companies can eliminate this uncertainty; that is why advisers tell investors to diversify their portfolios, and why companies do not pay cash when acquiring young, high-growth firms.

SUMMARY

The emergence of Internet, mobile, and other technology companies has created impressive value for some high-growth enterprises. It has also raised questions about the sanity of a stock market that at times has appeared to assign higher value to companies the more their losses mounted. But as this chapter demonstrates, the DCF approach remains an essential tool for understanding the value of high-growth companies. You must adapt your approach when valuing these companies: start from the future rather than the present when making your forecast, think in terms of scenarios, and compare the economics of the business model with peers. Though you cannot reduce the volatility of these companies, you can at least understand it.

REVIEW QUESTIONS

1. Which valuation methodology—scenario-based discounted cash flow analysis or multiples analysis—is preferable in valuing high-growth firms? Why?
2. How would you approach valuing a two-division firm with one mature division and one high-growth division?
3. How does the total market for a new product differ from a company's addressable market? Which market is more relevant for forecasting a company's revenues?
4. For a company with a new product, how can you estimate its potential market share?
5. How do you estimate the potential margin and capital turnover for a young, high-growth company? Are the company's current margin and capital turnover relevant?
6. Why do most young, high-growth companies have negative earnings?
7. Last year, GrowthCo traded at \$20 per share. Over the past 12 months, the company's share price rocketed to \$60 per share. Does this mean the share price was misvalued last year?